

OVERVIEW



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Three levels of framing

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Abstract

A sociologist and a linguist, unaware of each other's work, each assigned a technical meaning to the term *frame* around 1970, based on separate usages of the word *frame* from the 1950s. Each researcher instigated a theory of frame analysis. Over the following decades, the two approaches to framing became intertwined as followers of both Goffman and Fillmore studied metaphoric framing, examined factors affecting the communication of frames, and became particularly interested in politics and the mass media. Years later, many theorists complain about the fragmented field of frame studies. The paper suggests that some of the fragmentation can be resolved by recognizing the dual origins of framing studies, and classifying instances of framing in either the Goffman or the Fillmore tradition as occurring at the level of language, thought, or communication. These three levels are termed semantic framing, cognitive framing, and communicative framing.

This article is categorized under:

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cognitive linguistics, conceptual metaphor, Fillmore, frame semantics, Goffman

1 | INTRODUCTION

When you hear someone yell, “I’m being robbed,” several types of framing happen at once. First, the semantics of the word *rob* inform you that a perpetrator is involved, so you might look around for a robber. Second, your background knowledge tells you that the police will arrest robbers, so you may decide to call the police. Third, the victim is claiming the event is a robbery, rather than an instance of overcharging or charitable donation, for example, which affects your response. The semantic requirements of *rob*, the conceptual prerequisites for a concept of robbery, and the choice to describe the event as a robbery, all could be considered framing.

Given this array of interpretations, it is unsurprising that framing has been recognized in Communication Studies as a “fractured paradigm” (Entman, 1993, p. 51) that is “characterized by theoretical and empirical vagueness” (Scheufele, 1999, p. 103). In Linguistics, framing is a similarly disputed concept that several linguists have attempted to clarify (Fillmore, 2008; Petruck, 2022) and even reconcile with the approaches to framing in Communication Studies,

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Sociology, and other fields (Brugman et al., 2017; Burgers et al., 2016; Fillmore & Baker, 2009; Tannen, 1979). Researchers in Communication Studies and related fields have written reviews of their framing literature (Borah, 2011) and attempted to unify their own approaches to framing (Hallahan, 1999; Scheufele, 1999), but have done so without citing formative framing work in Linguistics and Cognitive Science such as Fillmore (1968) or Minsky (1974). An overview of significant work from the perspective of Linguistics and Cognitive Science is therefore overdue. A typology of frames from this perspective may not only help distinguish the various approaches to framing within Linguistics and Cognitive Science, but may fill in some of the informational gaps that have “fractured” framing studies.

Researchers in both Communication Studies and Linguistics seem largely unaware that the concept of frames arose independently (at least) twice: once in Sociology, leading to the approaches in Communication Studies and Media Studies (Goffman, 1974; inspired by Bateson, 1955); and once in Cognitive Psychology, evolving into the frames employed in Linguistics, Computer Science, and Artificial Intelligence (Fillmore, 1968; based on Bartlett, 1932). Subsequent work within both framing traditions has brought the two lines of research closer together, to the point where it is often forgotten that they had different origins. In particular, research on the persuasive power of metaphoric framing, as in the well-known work of linguist Lakoff (1996, 2004), has fed into both framing traditions and blurred the divide between them. This crossover between framing lineages, when added to the confusion of framing characterizations already present within each tradition, has created a tangle of definitions so dense and snarled that some theorists have thought it best to simply “abolish” frames (van Dijk, 2023, p. 172).

The current overview aims to improve communication among framing researchers by distinguishing three types of framing, explaining how these different approaches arose, and delineating the advantages and limitations of each. The three types of framing are here characterized as focusing on (1) language, (2) cognition, and (3) communication. These three types will be referred to as *semantic framing*, *cognitive framing*, and *communicative framing*, respectively. In Linguistics, the original “case frames” in Fillmore (1968) led to approaches that focus on the semantic requirements of individual words (that is, *semantic framing*), such as the fact that the word *rob* implies the existence of a perpetrator and a victim. However, some of the best-known framing studies in Linguistics, including the work of Lakoff, examine conceptual structures (i.e., *cognitive framing*). The cognitive frame for the concept of robbery, for example, assumes background knowledge including the existence of private property, illegal activity, and much more.

Even though the nature and usefulness of frames are hotly debated in Communication Studies and Media Studies (as summarized by Entman, 1993; Scheufele, 1999), Sociology and Political Science (van Dijk, 2023), all the disputed definitions of “frames” in these fields seem to have evolved from the sociological definition of frames set out by Goffman (1974). Most of these approaches focus on the communicative aspects of framing—such as the choice to present an event as a robbery rather than a misunderstanding—and are therefore here collectively called *communicative framing*. Many approaches in this tradition, including the original Goffman (1974), additionally pay close attention to conceptual aspects of framing (*cognitive framing*), so it is an oversimplification to associate Goffman with communicative frames and Lakoff with cognitive frames, for example. However, Goffman's followers in Sociology and allied fields have concerned themselves with communicative frames to a greater extent than Fillmore's followers in Linguistics, for instance, so the Goffman tradition can be meaningfully linked to a focus on communicative frames.

Though the definitions of framing by followers of Goffman are contentious and encompass various distinct subtypes, researchers are well aware of these divisions thanks to the work of scholars such as Entman (1993), Scheufele (1999), Borah (2011), and van Dijk (2023). The current paper therefore does not focus on the varied definitions and typologies of framing within Media Studies or Sociology, for example, but instead aims to delineate the differing definitions of framing within Linguistics and allied fields, as well as the divisions between these fields and those following the Goffman tradition.

To disentangle the three major levels of framing—semantic, cognitive, and communicative—this paper will begin (in Section 2) with a history of how the concept of “framing” developed from its two distinct origins into a plethora of overlapping phenomena. Section 3 describes researchers' attempts to bring logic to the resulting mess of frame types, usually by recognizing that there are two valid types of frames. For example, Fillmore (2008) distinguishes the frames here termed “semantic” and “cognitive,” and Entman (1993) recognizes the frames here called “cognitive” and “communicative.” However, it will be argued in Section 4 that three basic types of frames are needed—semantic, cognitive, and communicative—and each of these types will be described in turn. Section 5 will suggest that by disentangling the different types of frames, each can be precisely defined and its appropriate functions made clear.

2 | THE DUAL ORIGINS OF FRAMES

The concept of framing is more than “fractured,” as Entman (1993) describes the theory in Communication Studies. The concept was never unified in the first place, since framing seems to have been invented independently by (at least) two researchers in different fields. In Sociology and related fields, frames date back to the sociologist Goffman's volume *Frame Analysis* (1974), which itself cites the related use of the word *frame* in Bateson (1955).

In Linguistics, the concept of frames developed from Bartlett's (1932) schema theory in Cognitive Psychology (Fillmore, 1975, p. 123) and is generally traced back to Fillmore's 1968 and 1975 papers (e.g., in Lakoff, 2004, p. 121) or Minsky's 1975 paper (e.g., in Petruck, 2022; Ziem, 2015). The word *frame* itself was introduced into Linguistics in the form of *syntactic frames* (Fries, 1952), the usage that inspired the sense of *frame* in Fillmore (1968). Syntactic frames usually consist of sentences with a blank space representing a missing word. Specific words can then be inserted into the gap to test for the words' acceptability within that syntactic frame. Syntactic frames are still widely employed in Linguistics, but do not seem to be confused with the three types of frames considered here, and will not be discussed further.

Inspired by syntactic frames, Fillmore (1968) introduced *case frames*, that is, sentences with a blank space that can only be filled by words with a particular semantic role, or “case,” such as Agentive, Instrumental, and so on (Fillmore, 1968, p. 48). Fillmore initially listed six cases, though he suggested others and recognized that “additional cases will surely be needed” (1968, p. 46). Case frames remain influential in studies of argument structure (Busse, 2012; Levin & Rappaport Hovav, 2005). They also are the direct precursors of semantic frames (discussed in Section 4.1), which form the basis of the large-scale FrameNet and allied projects, for example (Ruppenhofer et al., 2016; FrameNet Brasil, 2023; FrameNet, 2023); as well as cognitive frames (explored in Section 4.2) as employed in best-selling books such as Lakoff (2004) and Luntz (2007).

The evolution from case frames to semantic and cognitive frames was encouraged by the work of mathematician and computer scientist Minsky (1974, 1975, 1980), who used the term *frame* for “a data-structure representing a stereotyped situation” (1975, p. 212). In a few brief pages, Minsky (1980) proposes that frames can model a wide variety of human abilities ranging from vision to grammaticality judgments. Most importantly for Linguistics, Minsky (1974, 1975, 1980) describes a type of frame very similar to Fillmore's later cognitive frames (see Section 4.2), and gives the example of a BIRTHDAY PARTY frame as a structure involving expectations such as gifts, cake, candles, and so on (this example is discussed further in Section 4.2). We can recognize a birthday party without speaking or otherwise communicating, so the frame is cognitive rather than semantic or communicative.

Although Fillmore and Minsky were writing about frames in the 60s and 70s, a different vision of framing was coming together in Sociology. The term *frame* as employed in Sociology, Communication Studies, and related fields is usually attributed to Goffman (1974), who defines the concept as the “principles of organization which govern events—at least social ones—and our subjective involvement in them” (1974, pp. 10–11). This definition would encompass a BIRTHDAY PARTY or many of the other cognitive frames discussed by Fillmore and Minsky. However, Goffman arrived at his use of the term *frame* by an entirely separate route. Fillmore for example, made it clear that his own research was in no way based on Goffman's. Fillmore's early work cites Goffman only in order to explain that his work is separate from Goffman (Fillmore 1975, p. 130). By 2008, Fillmore seems less vehement about the distinction, however, and he mentions Goffman as one of the researchers working on the social aspects of framing (Fillmore, 2008, p. 3; Fillmore & Baker, 2009, p. 315).

Goffman's frames (1974) were not based on syntactic frames (Fries, 1952), case frames (Fillmore, 1968), or Minsky's frames (1975). Instead, Goffman credits his use of the word *frame* to Bateson (1955), who assigns nonserious play activities to a play “frame” (1955, pp. 70–73). If certain activities can be “framed” as play, Goffman reasons, then other social events can be “framed” as theater or sport, for example. Crucially, Goffman's frames can be “transformed” by individuals in order to manipulate others, to generate humor, or for other purposes. Goffman gives the example of a car advertisement that frames driving as a fun leisure activity, by including symbols of leisure, such as a beach. The persuasive power of frames is the lynchpin of the structures here termed *communicative frames*, which consist of communicative strategies that prompt particular patterns of thinking.

The persuasive potential of frames also inspired linguists such as Reddy (1979), who first wrote about the power of metaphoric framing, and Lakoff, the author of the best-selling framing guide *Don't Think of an Elephant* (2004). Lakoff adopted the cognitive frames developed by Fillmore and Minsky, but integrated these into Conceptual Metaphor Theory (CMT), a model he expounded in another best-selling book, *Metaphors We Live By* (Lakoff & Johnson, 1980). CMT is founded on the premise that metaphors shape the way we think, and therefore the way we live, as suggested by the

1980 title. Lakoff follows Reddy (1979) in referring to these functions as *framing* (Lakoff, 1996; Lakoff & Johnson, 1980). Insofar as Reddy and Lakoff focus on the way that metaphors affect thought, their work deals with cognitive framing. Indeed, Lakoff frequently credits Fillmore as his inspiration, even referring to him as the “Father of Frame Semantics” (2004, p. 121). Lakoff’s best-selling books and public persona propelled him to a level of notoriety beyond that of Fillmore, however. In 2005, the *New York Times* even dubbed Lakoff the “father of framing” (Bai, 2005), giving Lakoff the title he had assigned to Fillmore the previous year.

Lakoff repeatedly attributes his frames to Fillmore’s tradition, and never cites Goffman in his work. Similarly, Goffman does not cite Lakoff’s work on frames, though he cites a paper co-authored by Lakoff on another topic (Goffman, 1976, p. 278). Despite the lack of interaction between Goffman and Lakoff, the latter has sometimes been mischaracterized as Goffman’s follower. Goffman is frequently mentioned in reviews of Lakoff’s work (Ohl et al., 2013; Williams, 2014), and Lakoff’s frame analysis is erroneously described as “based on the early work of sociologist Erving Goffman” (Gurman, 2009). The two scholars’ work seems to be associated in readers’ minds despite its lack of historical relatedness.

Though Lakoff did not claim to be working in Goffman’s tradition, Lakoff’s work arguably did bring the concept of frames in Linguistics closer to the cognitive and communicative frames in Sociology, Communication Studies, and related fields. From its onset, Lakoff and Johnson’s CMT (Lakoff & Johnson, 1980) focused on the ways that metaphor shapes our thinking (i.e., cognitive framing). Lakoff and Johnson claim that “the way we think, what we experience, and what we do every day is very much a matter of metaphor” (1980, p. 4). Increasingly, however, Lakoff became interested in the power of metaphor to affect the thinking of those around us (i.e., communicative framing). In *Moral Politics*, Lakoff’s framing guide for Democrats, he writes, “To use the other side’s words is to accept their framing of the issue” (1996, p. 410). When you speak or write, Lakoff recommends, you should choose words that support the cognitive frame that you want “the other side” to have. Lakoff’s take on framing as a persuasive tool resembles the view in Communication Studies, in which “(a)nalysis of frames illuminates the precise way in which influence over a human consciousness is exerted by the transfer (or communication) of information” (Entman, 1993, p. 51).

Whereas Lakoff’s interest in metaphor pulled him into the study of politics, some political researchers in the Goffman tradition have been dragged into metaphor studies. In a meta-analysis of 319 political framing experiments, Brugman et al. (2017) found that nearly one-third of the studies examined metaphorical framing, though the framing researchers themselves did not always recognize frames as metaphoric. Even Krippendorff (2017), who is critical of framing in general, acknowledges the importance of metaphor in communicative framing. “Metaphors are ... prototypical initiators of framing,” he writes (2017, p. 97). Metaphor has become an essential part of communicative framing studies in the Goffman tradition, just as some of the best-known conceptual metaphor studies in the Fillmore tradition, such as Lakoff (1996, 2004), focus on communicative framing.

The two traditions have been further united by linguistics researchers’ efforts (Matsumoto, 2010) to revive the “interactional frames” initially proposed by Fillmore (1982), which represent the conceptualizations of interactive contexts ranging from recipes to romantic dates. These cognitive frames model social interactions, much as Goffman’s original (1974) frames include conceptualizations of play, theater or sport, for example; so, the revitalization of interactional frames draws attention to a substantial area of overlap between Fillmore (1982) and Goffman (1974).

For the past three decades, then, communicative framing, metaphoric framing, and interactional framing have brought the Fillmore and Goffman traditions closer together—to the point where the two traditions have merged in the minds of some researchers (Gurman, 2009; Ohl et al., 2013; Williams, 2014).

3 | PRIOR DISTINCTIONS BETWEEN FRAME TYPES

Although some scholars have suggested abandoning frame analysis (Krippendorff, 2017; van Dijk, 2023), many more have tried to salvage it. This section summarizes the attempts to heal the “fractured paradigm” of framing (Entman, 1993). Typically, these attempts consist of defining and disentangling two types of frames that have previously been confused or conflated. The frame typologies introduced in this section will subsequently (in Section 4) be aligned and fitted into a more general model comprising semantic frames, cognitive frames, and communicative frames.

Fillmore himself recognizes the division between cognitive frames and semantic frames, though he calls the latter “linguistic frames” (Andor, 2010; Fillmore, 2008) or “linguistically anchored frames” (Fillmore & Baker, 2009). Fillmore clearly defines the two types in his 2008 book chapter “The Merging of ‘Frames’”, and redefines them to some extent the next year (Fillmore & Baker, 2009). However, his terminology for the types has not been consistent over the course

of his career. In earlier works, Fillmore sometimes discriminates the two by referring to cognitive frames as “scenes,” reserving “frames” for semantic frames (Fillmore, 1975, 1976, 1977). By 1985, Fillmore had abandoned this use of “scene” (Petruck, 2022). The distinction between cognitive frames and semantic frames remains implicit in his writing, but not overt in his choice of terminology, until 2008.

Like many linguists, Fillmore focuses on semantic and cognitive frames. However, the existence of communicative frames is implied in his work. Fillmore (1982) argues that “the same ‘facts’ can be presented within different framings, framings which make them out as different ‘facts’” (1982, p. 125). His examples include the contrast between the words *thrifty* and *stingy*, which describe the same antispending behavior but make different assumptions about values. That is, *thrifty* assumes that saving money is admirable and wastage is to be frowned upon, whereas *stingy* implies that spending money can be beneficial and unwillingness to do so can be a negative trait. By using someone else's choice of words, such as *stingy* rather than *thrifty*, you “accept their framing of the issue,” as Lakoff puts it (1996, p. 410). To reject the framing imposed by *stingy*, and the assumption that spending is advantageous, you must choose a word with different assumptions, such as *thrifty*. Fillmore's “different framings” therefore fall within the category here termed *communicative frames*.

Outside Linguistics, numerous frame researchers have distinguished the two types of frames here described as *cognitive frames* and *communicative frames*. For example, Kinder and Sanders (1990, p. 74; 1996, p. 164) compare frames that “are internal structures of the mind” versus frames that are “embedded in political discourse.” The “structures of the mind” are aspects of individual thought, that is, cognitive frames; whereas those “embedded in political discourse” are facets of communication (albeit specifically political communication, unlike the more general communicative frames suggested here). Entman (1993) differentiates the “information-processing schemata” within individual minds from the “attributes of the news itself” (1993, p. 7). That is, he distinguishes cognitive frames from the communicative frames that influence them. Later on, Scheufele (1999) adopts the two types in Kinder and Sanders (1990) but refers to “individual frames” rather than “structures of the mind,” and employs the term “media frames” instead of frames “embedded in political discourse” (1999, p. 103). Both Kinder and Sanders’ “political discourse” frames and Scheufele’s “media frames” refer to only a subset of the structures here considered communicative frames, namely those employed in political communication and the news media. Nonetheless, Scheufele, like Kinder and Sanders, finds it necessary to contrast the communicative “political” or “media” frames with another type equivalent to the cognitive frames discussed here.

Finally, Druckman (2001a) makes a distinction between “frames in thought” and “frames in communication,” which equate closely to the cognitive and communicative frames distinguished here. Interestingly, Druckman (2001a) considers both Goffman's and Fillmore's frames as “frames in thought” (pp. 227–228), despite the different histories of these traditions (see Section 2). Indeed, Goffman (1974) seems primarily concerned with cognitive frames, despite many of his followers' focus on communicative frames.

Fillmore's “linguistic frames,” here termed *semantic frames*, seem distinct from any of the frames identified by Druckman. Of course, most researchers focusing on cognitive frames or communicative frames are aware of the role of language in these types of framing. For example, Entman writes that communicative frames “are manifested by the presence or absence of certain key words, stock phrases, stereotyped images, sources of information, and sentences” (1993, p. 52). Similarly, van Gorp (2010) distinguishes between cognitive frames, which he terms a “reasoning device,” and the words that *evoke* cognitive frames in the sense of Fillmore (2008, p. 2), which van Gorp calls “lexical choices” within the “framing device” (van Gorp, 2010, p. 35).

Recognizing the importance of language in framing, however, does not constitute semantic framing in the sense used here, which refers to the semantic requirements of individual linguistic units. Researchers in Linguistics, Artificial Intelligence, and allied fields have developed methods of extracting these semantic requirements from large corpora (collections) of spoken and/or written language, and established formalisms for representing the resultant semantic frames (see Section 4.2). Acknowledgement of the role of “words” (Entman, 1993), and “lexical choices” (van Gorp, 2010) does not constitute semantic framing in this sense. On the other hand, researchers in fields like Communication Studies have considered more factors affecting the communication of frames than linguists or cognitive scientists, for example. Putting these types of frames together into a unified model is therefore necessarily an interdisciplinary effort.

4 | THREE FRAME TYPES

Linguists, sociologists and others have succeeded in extricating semantic frames from cognitive frames; and cognitive frames from communicative frames. In general, Fillmore and his followers have focused on semantic and cognitive

frames, whereas Goffman's followers have been interested in cognitive and communicative frames. However, when the efforts of both traditions are compared, it becomes apparent that they describe three distinct types of frames: semantic, cognitive, and communicative, operating at the levels of language, thought, and communication, respectively. Though these three levels have not previously been distinguished for frames, it should be noted that Steen (2008) suggests three levels of metaphor that are analogous to the three levels of framing introduced here, relating to “language, thought, and communication” (2008, p. 214). When a phenomenon—such as metaphor or framing—exists in cognition but can be communicated to others, these three levels are a natural consequence. We have to be able to think about something in order to communicate about it, and of course, much of our communication involves language. Cognitive structures will therefore often develop communicative and linguistic counterparts.

Of the three types of frames, semantic frames were the first to be called *frames*, insofar as they are the direct descendants of Fillmore's case frames (Fillmore, 1968). The theory and practice of semantic frames continues to evolve, as reflected by the projects and events listed by Global FrameNet, for example (FrameNet Brasil, 2023). Generally speaking, semantic frames specify the semantic requirements of words and other linguistic forms (FrameNet, 2023; Ruppenhofer et al., 2016, as discussed in Section 4.1). In contrast, cognitive frames are open-ended structures consisting of the background knowledge necessary to understand a concept, as described in Section 4.2. Finally, communicative frames arise when cognitive frames are encoded in language, art, or another medium, causing the cognitive frames to be created or activated in someone else's mind (explored in Section 4.3).

These levels of framing are not only interrelated; some types are existentially dependent on others. For example, communicative frames could not exist without cognitive frames, since cognitive frames are the mechanisms by which communicative frames influence thought. Semantic frames also rely on cognitive frames. Without an underlying conceptual organization of experience, semantic frames that require specific aspects of this experience to be encoded could not occur. Although communicative frames do not require semantic frames (and can involve other communicative resources such as art), in practice most communicative frames include language, and therefore semantic frames.

Conversely, cognitive frames do not require either of the other types. We can recognize the signs of a BIRTHDAY PARTY (nicely dressed children, cake, balloons) without talking about it or otherwise communicating the frame to anyone else. Similarly, semantic frames do not require communicative frames, since we can employ language in thought, speech, or writing when we have no intention of communicating.

The fact that cognitive frames underlie each of the other types helps explain why cognitive frames have received attention in the widest variety of fields, ranging from Linguistics to Political Science. Several researchers who study semantic frames or communicative frames have noticed that the phenomena that they study rely on cognitive frames, and have observed that the analysis of semantic or communicative frames can benefit from the study of cognitive frames. Realizations of this kind inspired many of the studies overviewed in section 3, in which researchers recognized both semantic and cognitive frames, or both communicative and cognitive frames.

This section turns, now, to an overview of the three framing types.

4.1 | Semantic frames

Of the three types of frames differentiated here, semantic frames are the most precisely defined and the most highly formalized. Though Fillmore (2008) chooses the label “linguistic frames” for these structures, the current paper employs *semantic frames* rather than “linguistic frames” for three reasons. Firstly, “linguistic frames” is unclear, because other frames are recognized in Linguistics, including cognitive frames (see Section 4.2) and the syntactic frames that gave rise to both cognitive and semantic frames (Fillmore, 1968). Secondly, the field of Semantics is the study of meaning in language, so the modifier “semantic” conveys the fact that semantic frames specifically model linguistic meaning. Finally, *semantic frames* highlights that these structures serve some of the same explanatory functions as semantic roles (also called participant roles [Allan, 1986], thematic roles [Dowty, 1991; Jackendoff, 1990], or theta roles [Chomsky, 1981]) such as *agent* and *patient*.

Semantic frames, as employed in the FrameNet project, are tightly focused on the semantic requirements of words. A frame is defined in *FrameNet II: Extended Theory and Practice* as “a script-like conceptual structure that describes a particular type of situation, object, or event along with its participants and props” (Ruppenhofer et al., 2016, p. 7). The “participants and props” are frame *elements* which consist of frame *roles* and their *fillers*. For example, the KILLING frame has roles such as KILLER OR CAUSE and VICTIM, because someone or something has to do the killing and someone or something must be killed. Each time a frame occurs, its roles may have different fillers. For example, in the sentence *Spot*

killed a mouse, the KILLER role is filled by *Spot* and the VICTIM role is filled by *a mouse*. In this sentence, the verb *kill* evokes the KILL frame. That is, when we encounter the verb *kill* in a sentence, we know that we need the KILLING frame to understand the sentence, and we know to expect other words in the sentence that refer to a KILLER or CAUSE and a VICTIM, along with other potential roles such as an INSTRUMENT, since these roles often occur in sentences about KILLING.

The roles that we expect when we encounter a semantic frame such as KILLING can be extracted from language data. The various FrameNets and related projects collect instances of frames such as KILLING from large text collections (corpora). For example, the original FrameNet project uses the 100-million-word British National Corpus, which was the largest balanced corpus of English available when the FrameNet project started in 1997. Over 200,000 sentences from this corpus were manually examined and searched for recurrent elements in grammatical relations to frame-evoking words such as *kill*. The elements in a frame consist of the words that are recurrently related to the frame-evoking word, such as the KILLER and the VICTIM that frequently occur in sentences with the verb *kill* or other words that evoke KILLING, such as *annihilate*, *annihilation*, *asphyxiate*, and 67 other words and phrases listed in the freely available FrameNet data (FrameNet, 2023).

The reference to “participants and props” in the definition of frames in Ruppenhofer et al. (2016, p. 7) also indicates that some frame roles are more essential to the frame than others, since “participants” are an essential part of an event whereas “props” are optional extras. The essential and optional roles in FrameNet are referred to as *core* and *non-core*, respectively. Core frame elements are “conceptually necessary” to the frame. The core elements of predicates look somewhat like Fillmore’s original case frames (Fillmore, 1968) or related concepts such as thematic roles (Allen, 1986; Chomsky, 1981; Dowty, 1991; Jackendoff, 1990).

Frame roles differ from thematic roles mainly in that they are often specific to the frame in question. For example, the verbs *eat* and *kill* both require the same thematic roles, an agent and a patient; whereas in FrameNet, *eat* evokes INGESTION with the core elements INGESTIBLES and INGESTOR, and *kill* evokes KILLING with a KILLER and a VICTIM, along with other potential core elements. In comparison with thematic roles, then, the FrameNet approach sacrifices generality for accuracy. For example, the label INGESTIBLES as opposed to *patient* recognizes that the selection restrictions of *eat* bar participants such as algorithms or black holes, which generally cannot be eaten but may be the “patients” of other predicates such as *calculate* or *identify*.

Finally, semantic frames are critical to the functioning of most metaphoric language. A word evokes the same semantic frame whether it is used metaphorically or non-metaphorically (Sullivan, 2006, 2013, 2017), which helps demonstrate how metaphoric language is similar to non-metaphoric language. Applying frame semantic methods to metaphoric language facilitates the automatic recognition and analysis of metaphors in a text corpus, for example (Dodge et al., 2015; Lederer, 2016; Sullivan, 2022).

This section has barely scratched the surface of the work on semantic frames, but has indicated that semantic framing research requires a precise, empirical, linguistic approach. Of course, semantic frames do not exist in isolation; they depend on cognitive frames and contribute to communicative frames.

4.2 | Cognitive frames

The ability to think is a prerequisite for language and communication; and cognitive frames must exist in order for either semantic frames or communicative frames to be possible. Cognitive frames have therefore caught the attention of researchers who are otherwise interested primarily in semantic or communicative frames. For the same reason, however, cognitive frames have been confused with both of these other types (see Section 3).

The “cognitive frames” explicitly recognized in Fillmore’s later work (Andor, 2010; Fillmore, 2008; Fillmore & Baker, 2009) are probably the best-known type of frames in Linguistics, since these appear in Lakoff’s work (1996) and some of Fillmore’s seminal publications (1982), though Fillmore does not call them “cognitive frames” until much later (see Section 3). In 1982, Fillmore defines a “frame” of this type as:

any system of concepts related in such a way that to understand any one of them you have to understand the whole structure in which it fits; when one of the things in such a structure is introduced into a text, or into a conversation, all of the others are automatically made available. (Fillmore, 1982, p.111)

For example, if a *birthday party* is mentioned, other concepts such as birthday presents and cake are made available (as discussed in Minsky, 1974, 1975, 1980). If we are invited to a birthday party, we may consider purchasing and gift-

wrapping a present, bringing a card, dressing nicely, and congratulating the person having a birthday. These are all elements of the cognitive frame of BIRTHDAY PARTY. However, a list of these elements will not be nearly enough to explain a birthday party to a person from a different cultural and linguistic background. The person would need to understand the solar calendar and keeping track of age in terms of years; the concept of material property and exchange that allows gifts to be purchased and given; and so forth. These additional requirements belong to the BIRTHDAY PARTY cognitive frame, but are inherited from more basic cognitive frames such as SOLAR CALENDAR, FINANCIAL TRANSACTION, and GIVING. We can think about the solar calendar without considering birthdays or birthday parties, but the reverse is not true, so SOLAR CALENDAR is inherited by BIRTHDAY PARTY but not the other way around. We need a lot of basic frames to understand a high-level frame such as BIRTHDAY PARTY.

Moreover, to understand the BIRTHDAY PARTY frame, we need to know what the *prototypical* birthday party is like. Prototype theory, as developed by Rosch (1973), Berlin and Kay (1969), and others, plays an important role in cognitive frames. The prototypical birthday party in Western culture probably celebrates a child or young person, and involves gifts of toys, party hats, balloons, cake, and lots of guests. The prototypical birthday is not the most frequent kind of birthday celebration, but is nevertheless the model we use when reasoning about the BIRTHDAY PARTY frame. As Fillmore (1982) observes, “often the frame or background against which the meaning of a word is defined and understood is a fairly large slice of the surrounding culture, and this background understanding is best understood as a ‘prototype’ rather than as a genuine body of assumptions about what the world is like” (1982, pp. 117–118). That is, we will not expect that every birthday party will be like the prototype, but we are still aware of it and can make reference to it. If someone celebrates a particularly low-key birthday, we might joke, “What, no balloons?” in reference to the prototype.

The BIRTHDAY PARTY frame is not linguistic knowledge per se, since it affects how people think and behave even when no language is involved. People who know the BIRTHDAY PARTY frame will recognize a birthday party even if nobody says *birthday* or *balloons* or *cake*. Cognitive frames, such as BIRTHDAY PARTY and the frames that it inherits, are *encyclopedic knowledge* in the sense of Kecskes (2013). Words such as *birthday* offer access to a vast encyclopedia of information structured by cognitive frames. These frames make it easier to communicate, because we can refer to the shared structures. However, we use them even when we are not speaking or communicating, such as when we are thinking about the time of year (SOLAR CALENDAR) or planning a child's birthday (BIRTHDAY PARTY), for example.

Cognitive frames are essential to language, since we rely on them to understand words such as *birthday* or *year*, or indeed most other words. They are also necessary background for semantic frames. In the KILLING frame, as discussed in the previous section, a word such as *kill* that evokes KILLING needs a semantic argument that fills the role of VICTIM. This requirement only makes sense because we have pre-existing cognitive frames related to the BODY, DEATH, CAUSATION, and MURDER, for example. We have to understand cognitively that living beings can die when their bodies are damaged in order to have any idea of what *kill* means in the first place, or to understand the role that a VICTIM has in KILLING, or to know that a victim will subsequently be dead after the killing has taken place, and so on.

In an interview near the end of Fillmore's life, he explains the relation between cognitive frames and semantic (“linguistic”) frames (Andor, 2010, p. 158):

(C)ognitive frames are those background understandings needed for making sense of things that happen around us, and linguistic frames are those that are specifically coded in—or “evoked by”—lexical units or other features of linguistic form. (Andor, 2010, p. 158)

To take a well-known example, the cognitive frame RESTAURANT provides a background understanding of how restaurants function. The cognitive frame informs us that if we enter a restaurant, we receive a menu, we place an order with the waitstaff, and we are brought food which we are then expected to pay for. In contrast, a semantic frame consists of the information required by a specific word such as *pay*, for example. When we use *pay* in a sentence, we normally include words that designate who is doing the paying, who received the payment, and the amount paid, as in *John paid the waiter \$30*. Clearly, cognitive frames potentially include a lot more information than semantic frames. The RESTAURANT frame contains everything we know about restaurants (such as that they hire waiters) plus the background knowledge that we need to understand these concepts (such as that people work at jobs to earn money). On the other hand, the semantic frame evoked by *pay* tells us only what kinds of words can go into a sentence along with the verb *pay*. There is no absolute limit to how much information a cognitive frame can hold; if a new trend (such as taking a birthday selfie each year) becomes part of the prototypical BIRTHDAY PARTY, this information and all its underlying cognitive frames will feed into BIRTHDAY PARTY, which will then contain more structure. The content of semantic frames is more circumscribed. A quick perusal of FrameNet's online data shows that semantic frames typically have 2–5 core elements

and a dozen or so noncore elements (FrameNet, 2023). Semantic frames acquire and lose elements over time as part of semantic change, but there appears to be a natural upper limit on their complexity that is not shared by cognitive frames.

Cognitive frames also differ from semantic frames in that they are activated by a wide range of thoughts, experiences, and language. Thinking about a menu, seeing a menu, or hearing the word *menu* will all evoke the *RESTAURANT* frame. In contrast, semantic frames are evoked only by units of language, such as the word *pay*.

In sum, cognitive frames are potentially more complex than semantic frames; can be evoked in more ways; and must be present in order for semantic frames to exist. The two types of frames each have their own characteristics; operate at different levels; and offer insights into distinct facets of language and thought.

4.3 | Communicative frames

Many of our thoughts can be communicated through language, art, music or some other means. When we communicate a cognitive frame to other people, they may activate that frame as a result. The relevant structure of the cognitive frame(s) selected for communication, plus all the factors that help determine whether and how the frame is communicated, constitute a *communicative frame* in the sense considered here. The structure of one or more cognitive frames is therefore subsumed in each communicative frame, much as each semantic frame includes only a subset of the structure in a cognitive frame. Communicative frames are therefore the most complex type of frame, followed by cognitive frames, trailed by semantic frames.

Entman defines this type of frame in the following way:

To frame is to select some aspects of a perceived reality and make them more salient in a communicating text, in such a way as to promote a particular problem definition, causal interpretation, moral evaluation, and/or treatment recommendation for the item described. (Entman, 1993, p. 52).

The “aspects of the perceived reality” are the structures here called cognitive frames. Many factors, however, can affect whether and how a cognitive frame is communicated. As Entman observes, “the presence of frames in the text, as detected by researchers, does not guarantee their influence in audience thinking” (Entman, 2003, p. 53). Cognitive frames in the mind and language of a speaker/writer do not necessarily result in communicative frames. The reasons behind this failure or success are complicated, involving the repetition of the frame, the identity of the participants, the medium of communication, the historico-political context of the communicative event, and so forth. Therefore, communicative frames encompass a wide range of information. This probably contributes to the relevance of these frames across a wide range of fields, from Peace Studies to Economics, but also renders these frames hard to define and increases the contentiousness of what constitutes such a communicative frame.

Though it is not possible here to discuss all the factors affecting communication, it is worthwhile to overview a few aspects of communicative frames that have been noted by previous researchers and that set these frames apart from cognitive frames. An attempt at communicating may be more or less successful, as Entman’s quotation above suggests; and its relative success can be determined by attributes of communicative frames that are not part of the cognitive frames. To consider a simple example, a cognitive structure may be made more salient when it is encountered often, such as when it is frequently repeated in the news media. This phenomenon has been called “agenda-setting” (Iyengar & Kinder, 1987). As Scheufele (2000) describes the phenomenon, “Mass media can influence the salience of certain issues as perceived by the audience; that is, the ease with which these issues can be retrieved from memory” (Scheufele, 2000, p. 300). A repeated frame will not only be evoked more often when it is encountered in the media, but also subsequently in other situations where it otherwise might not be evoked. The fact that a cognitive frame can become more salient through repetition, for example, demonstrates that the salience of a communicative frame does not depend solely on the structure of the cognitive frame(s) that underlie it.

Increasing the salience of a frame can result in “priming,” a term borrowed from Psychology (Collins & Loftus, 1975; Tulving & Watkins, 1975). In Media Studies, “priming” refers to the frame’s effects on the audience’s opinions, particularly regarding politicians or political parties. As Scheufele (2000) describes the priming effect, “perceived issue salience ... influences the role that these issues or considerations play when an individual makes a judgment about a political actor” (Scheufele, 2000, p. 300). The political consequences of frame salience, like the salience of frames themselves, are not part of cognitive frames. Some communication theorists argue that agenda setting and priming

should not be considered “framing” (Borah, 2011; Cappella & Jamieson, 1997). Here it is suggested that while they play no role in cognitive framing, agenda setting and priming can be treated as aspects of communicative framing since they influence the communication of frames.

Many other factors shape the salience of communicative frames. For example, audiences are more likely to activate a frame when it comes from a trusted source (Druckman, 2001b; Miller & Krosnick, 2000). Frames that are communicated visually, for example, will have different effects than frames that are encoded in language, or which are expressed both visually and linguistically (Edell & Staelin, 1983; Kamins & Marks, 1987).

Though it is useful to recognize subcategories of communicative framing, this project is outside the scope of the current paper. Subtypes of communicative framing are proposed or overviewed in Hallahan (1999), Scheufele (2000), Snow et al. (2007), Borah (2011) and van Dijk (2023), for example. Communicative framing has been subjected to substantial meta-analysis that has largely bypassed semantic and cognitive framing. When Entman calls framing a “fractured paradigm” (1993, p. 51), he assumes that frames “invariably involve communication” (1993, p. 51), demonstrating that he is interested only in communicative frames. Though van Dijk (2023, p. 154) recognizes that framing has been used to “refer to a large variety of discursive and cognitive phenomena,” his discussion focuses on communicative frames, touches on cognitive frames, and ignores semantic frames.

The aim of the current paper is not to reinvent the wheel by proposing subcategories of communicative frames. Rather, the current goal is to step back, view communicative framing in a larger context, and show that all types of communicative frames are more closely related to each other than they are to cognitive or semantic frames.

5 | CONCLUSION

The decades of confusion between types of framing have done little to advance the study of frames in any field. The convoluted history of frame studies has arguably slowed progress in framing research of all types. When linguists, sociologists, or other framing researchers encounter an approach with different assumptions and methods, their first reaction may be to reject the work as baseless and misguided. As a result, advances in framing research made using one approach are less likely to benefit framing research in another area. Even worse, flawed assumptions or procedures that are corrected in one approach are likely to persist in others. As Entman eloquently puts it in his 1993 critique of the “fractured paradigm” of framing, “hypotheses thoroughly discredited in one field may receive wide acceptance in another” (1993, p. 51).

The one aspect of frame studies that everyone seems to agree on is that multiple conflicting definitions of frames exist. One reaction to this state of affairs would be to “abandon using these notions” as van Dijk (2023, p. 154) recommends to researchers in his field, Social Movement Research. However, if researchers were to “abolish” frames, as van Dijk (2023, p. 174) and Krippendorff (2017) propose, a great deal of promising work would be lost. For example, semantic frame research has led to FrameNet projects around the world in numerous languages (e.g., Boas, 2009; Lönneker-Rodman, 2008; FrameNet Brasil, 2023; Yang et al., 2018) and inspired the MetaNet project collecting and analyzing metaphors (Dodge et al., 2015; Petruck, 2016). Researchers who decry the degradation of communicative framing studies, such as Krippendorff (2017), may not even be aware of the work on semantic frames taking place outside their own area of expertise.

Throughout the history of frames studies, innovative and important work has persisted amidst the misunderstandings and unclear terminology. Although some branches of frame studies have fallen away, others have grown more vibrant, and promise to yield insights regarding language, metaphor, and even human nature. Recognizing the different origins of these branches helps explain the current diversity in the field. Understanding that the branches relate to different levels of framing may help untangle the work within framing studies; improve communication between disciplines; and ensure that productive framing research is not discarded on the basis of its connection, or lack thereof, to work at a different level of framing.

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Karen Sullivan: Conceptualization (equal); formal analysis (equal); investigation (equal); methodology (equal); writing—original draft (equal); writing—review and editing (equal).

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